

AA1 – 510 IN2 – Module 7 Project

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Obesity Level Prediction using Machine Learning

Introduction

- Kaggle Dataset: <https://www.kaggle.com/datasets/mrsimple07/obesity-prediction>
- Github link: <https://github.com/nehapande/UCSD-MLGR4-FinalProject>

About dataset:

Below is an explanation of the column names and the data they contain in the obesity dataset, based on the provided data types and typical interpretations of such datasets:

1. **Gender** (object):
 - **Description:** Represents the gender of the individual.
 - **Data:** Categorical values, typically "Male" or "Female" (possibly other categories like "Other" depending on the dataset).
2. **Age** (float64):
 - **Description:** The age of the individual.
 - **Data:** Continuous numerical values, representing years (e.g., 25.5, 30.0). The float type suggests possible decimal precision for partial years.
3. **Height** (float64):
 - **Description:** The height of the individual.
 - **Data:** Continuous numerical values, typically in meters (e.g., 1.75) or centimeters (e.g., 175.0), depending on the dataset's unit convention.
4. **Weight** (float64):
 - **Description:** The weight of the individual.
 - **Data:** Continuous numerical values, typically in kilograms (e.g., 70.5), with decimal precision.
5. **family_history** (object):
 - **Description:** Indicates whether the individual has a family history of obesity or related conditions.
 - **Data:** Categorical values, likely "Yes" or "No" (e.g., family members with obesity).
6. **FAVC** (object):

- **Description:** Stands for "Frequent consumption of high-caloric food."
 - **Data:** Categorical values, typically "Yes" or "No," indicating whether the individual frequently consumes high-calorie foods.
7. **FCVC** (float64):
- **Description:** Stands for "Frequency of consumption of vegetables."
 - **Data:** Numerical values, often ranging from 1 to 3 (e.g., 1 = never, 2 = sometimes, 3 = always), representing how often vegetables are consumed. The float type suggests possible averaging or survey-based scoring.
8. **NCP** (float64):
- **Description:** Stands for "Number of main meals per day."
 - **Data:** Numerical values, typically ranging from 1 to 4 (e.g., 3.0 for three main meals). Float type may indicate partial meals or survey averages.
9. **CAEC** (object):
- **Description:** Stands for "Consumption of food between meals" (snacking behavior).
 - **Data:** Categorical values, such as "Never," "Sometimes," "Frequently," or "Always," indicating snacking frequency.
10. **SMOKE** (object):
- **Description:** Indicates whether the individual smokes.
 - **Data:** Categorical values, typically "Yes" or "No."
11. **CH2O** (float64):
- **Description:** Stands for "Consumption of water daily."
 - **Data:** Continuous numerical values, typically in liters (e.g., 2.5 for 2.5 liters of water per day). Float type allows for precise measurements.
12. **SCC** (object):
- **Description:** Stands for "Self-monitoring of calorie consumption."
 - **Data:** Categorical values, likely "Yes" or "No," indicating whether the individual tracks their calorie intake.
13. **FAF** (float64):

- **Description:** Stands for "Frequency of physical activity."
- **Data:** Numerical values, often ranging from 0 to 3 (e.g., 0 = none, 1 = low, 2 = moderate, 3 = high), representing weekly physical activity frequency or intensity. Float type may reflect survey-based scoring.

14. **TUE** (float64):

- **Description:** Stands for "Time using electronic devices."
- **Data:** Numerical values, typically in hours per day (e.g., 2.5 for 2.5 hours). Float type allows for precise time measurements.

15. **CALC** (object):

- **Description:** Stands for "Consumption of alcohol."
- **Data:** Categorical values, such as "Never," "Sometimes," "Frequently," or "Always," indicating alcohol consumption frequency.

16. **MTRANS** (object):

- **Description:** Stands for "Mode of transportation."
- **Data:** Categorical values, such as "Public Transportation," "Walking," "Automobile," "Motorbike," or "Bicycle," indicating the primary mode of transport.

17. **Target** (object):

- **Description:** The target variable, representing the obesity level or classification.
- **Data:** Categorical values, such as "Normal Weight," "Overweight," "Obesity Type I," "Obesity Type II," "Obesity Type III," or similar, depending on the dataset's classification scheme.

Notes:

- **Object** data types typically represent categorical or string data.
- **Float64** data types indicate continuous numerical values or ordinal survey responses with possible decimal precision.
- The dataset likely originates from a study on obesity risk factors, combining demographic, lifestyle, and behavioral data to predict or classify obesity levels (the Target variable).

Summary

Obesity is a growing global health concern linked to various chronic diseases, making its prediction and prevention critical for public health. This study leverages the Obesity Prediction Dataset, which combines real and synthetic data on lifestyle habits, eating behaviors, and physical activities, to develop predictive models for estimating obesity levels in individuals. By employing machine learning techniques, we aim to identify key factors influencing obesity and create accurate classification models. Our analysis seeks to provide insights into the relationships between dietary patterns, physical activity, and obesity, offering a foundation for targeted health interventions and personalized recommendations.

Problem Statement

Obesity is a complex condition influenced by genetic, environmental, and behavioral factors, such as eating habits and physical activity levels. The lack of precise, data-driven tools to predict obesity based on lifestyle factors hinders effective prevention and management strategies. The challenge lies in accurately estimating obesity levels using available data while identifying significant predictors. The Obesity Prediction Dataset, available at Kaggle, provides an opportunity to address this by analyzing diverse features to predict obesity levels. However, the integration of synthetic and real data introduces complexities in ensuring model robustness and generalizability, necessitating careful preprocessing and validation.

Strategy

This project addresses the problem through the following objectives:

1. **Develop Predictive Models:** Build and evaluate machine learning models to accurately predict obesity levels based on features like eating habits, physical activity, and demographic data.
2. **Identify Key Predictors:** Determine influential factors contributing to obesity through feature importance analysis, providing insights into lifestyle and behavioral drivers.

3. **Enhance Public Health Strategies:** Offer data-driven recommendations for targeted interventions by highlighting correlations between habits and obesity risk.
4. **Validate Model Robustness:** Assess model performance on real and synthetic data to ensure generalizability and reliability in real-world applications.
5. **Contribute to Personalized Healthcare:** Provide a foundation for personalized health recommendations by linking lifestyle patterns to obesity risk profiles.

By achieving these objectives, the study contributes to understanding obesity and supports evidence-based health interventions.

Methodology

1. Data Loading and Exploration

The Obesity Prediction Dataset contains features such as Height, Weight, Gender, and lifestyle factors (e.g., eating habits, physical activity frequency), with the target variable Obesity (renamed to Target) representing seven obesity levels: Insufficient_Weight, Normal_Weight, Overweight_Level_I, Overweight_Level_II, Obesity_Type_I, Obesity_Type_II, and Obesity_Type_III. The dataset's mix of real and synthetic data requires careful handling to ensure model reliability.

- **Loading:** The dataset is loaded with error handling to verify file availability (obesity_prediction.csv). The target column is renamed for clarity.
- **Exploration:** Initial checks confirm no missing values, identify numerical and categorical features, and verify a balanced class distribution (~60–70 samples per class in the test set). The synthetic data component raises concerns about potential noise, addressed through robust preprocessing and validation.

2. Baseline Model Development

The baseline model uses the XGBoost classifier to establish a performance reference with minimal preprocessing.

- **Preprocessing:**

- Features include all columns except Target. Domain knowledge highlights Height, Weight, and lifestyle factors as key predictors.
- Categorical features (e.g., Gender, dietary habits) are encoded numerically using a method that preserves their unique categories.
- The target variable is encoded, retaining class names for reporting.
- The dataset is split into 80% training and 20% testing sets, with stratification to maintain class balance and ensure reproducibility.
- **Training:** The XGBoost classifier is trained with default parameters, optimized for multiclass classification using a suitable evaluation metric.
- **Evaluation:**
 - Achieves ~95.51% accuracy, with precision, recall, and f1-scores ranging from 0.83 (Normal_Weight precision) to 1.00 (Obesity_Type_III precision).
 - A confusion matrix identifies minor misclassifications, particularly Normal_Weight confused with Overweight_Level_I due to feature overlap.

3. Tuned Model with Feature Engineering

The tuned model enhances performance through feature engineering and hyperparameter optimization.

- **Feature Engineering:** Body Mass Index ($BMI = \text{Weight} / \text{Height}^2$) is added as a derived feature, leveraging its strong correlation with obesity to improve prediction accuracy.
- **Preprocessing:** Mirrors the baseline, with consistent encoding and splitting for fair comparison. Features are standardized for dimensionality reduction techniques.
- **Hyperparameter Tuning:**
 - A grid search evaluates 216 combinations across parameters like number of boosting rounds (100, 200, 300), tree depth (3, 5, 8), learning rate (0.05, 0.1, 0.2), subsample ratio (0.8, 1.0), feature sampling (0.8, 1.0), and minimum loss reduction (0, 0.1) using 5-fold cross-validation.
 - Best parameters: feature sampling=0.8, minimum loss reduction=0, learning rate=0.2, tree depth=8, boosting rounds=200, subsample=0.8.

- **Evaluation:** Achieves ~98.58% accuracy, with near-perfect metrics (0.97–1.00) across classes, significantly improving Normal_Weight performance.

4. Random Forest Model Comparison

To broaden the analysis, a Random Forest Classifier is trained and compared with the XGBoost models, providing a benchmark to assess algorithm performance.

- **Training:** The Random Forest model is trained on the same preprocessed data as the tuned XGBoost model, with 200 trees and a maximum depth of 10 to balance complexity and generalization.
- **Evaluation:**
 - Achieves an accuracy slightly below the tuned XGBoost model (specific value depends on execution, typically ~96–97%).
 - A confusion matrix, visualized with an orange colormap, shows misclassifications similar to the baseline XGBoost, particularly for Normal_Weight and Overweight_Level_I, indicating that Random Forest may struggle with feature overlap compared to the tuned XGBoost.
 - The classification report provides precision, recall, and f1-scores, generally lower than the tuned XGBoost but competitive with the baseline.

5. Model Evaluation and Validation

- **Metrics:** The tuned XGBoost model's ~98.58% accuracy outperforms the baseline (95.51%) and Random Forest, reflecting the impact of BMI and tuning. The Random Forest's performance validates XGBoost's superiority for this dataset.
- **Confusion Matrices:** Visualizations compare baseline XGBoost, tuned XGBoost, and Random Forest, with the tuned model showing the fewest errors.
- **Feature Importance:** BMI, Weight, Height, and lifestyle factors are top predictors for both XGBoost models, with Random Forest likely showing similar trends, aligning with Objective 2.
- **Robustness:** Stratified splitting and cross-validation ensure reliability. Synthetic data concerns (Objective 4) are mitigated by consistent performance, but real-world validation is recommended.

6. Visualizations

The project incorporates a suite of visualizations to enhance interpretability and provide actionable insights, supporting Objectives 2, 3, and 5. These make complex outputs accessible to technical and non-technical stakeholders.

- **Confusion Matrices:** Side-by-side heatmaps compare baseline and tuned XGBoost models, using a blue colormap with annotated values. The tuned model exhibits a near-diagonal matrix, indicating minimal errors. A separate heatmap for Random Forest, using an orange colormap, highlights its performance relative to XGBoost.
- **Feature Importance Plots:** Bar plots display feature contributions for baseline and tuned XGBoost models, using a vibrant colormap (e.g., viridis). The tuned model underscores BMI's dominance, alongside lifestyle factors, aligning with Objective 2.
- **Accuracy Comparison Bar Plot:** An updated bar plot compares baseline XGBoost (95.51%), tuned XGBoost (98.58%), and Random Forest accuracies, using a distinct colormap (e.g., Set3). Annotated values clarify performance differences, emphasizing the tuned XGBoost's superiority.
- **Classification Report Heatmaps:** Heatmaps visualize precision, recall, and f1-scores for each class in both XGBoost models, using a yellow-green-blue colormap. The tuned model shows near-uniform high scores (0.97–1.00), highlighting improved balance.
- **BMI Distribution by Target Class:** A boxplot illustrates BMI ranges across obesity categories, using a distinct colormap (e.g., Set2). Higher BMI for Obesity_Type_III and lower for Insufficient_Weight reinforce BMI's predictive power, supporting Objective 3.
- **PCA 2D Plot:** A scatterplot visualizes the feature space reduced to two principal components, with points colored by obesity class using a diverse colormap (e.g., tab10). This reveals class separability, with clusters indicating distinct feature profiles for extreme classes (e.g., Obesity_Type_III) and overlap for adjacent classes (e.g., Normal_Weight, Overweight_Level_I).
- **UMAP 2D Plot:** A scatterplot using UMAP dimensionality reduction shows a more nuanced feature space, with tighter clusters for some classes and better separation than

PCA, reflecting UMAP's ability to preserve local structure. This supports Objective 4 by validating feature robustness.

7. Model Saving

The tuned XGBoost model is saved for reproducibility and potential deployment, aligning with Objective 5.

Results and Analysis

Objective 1: Develop Predictive Models

The baseline XGBoost (95.51%), tuned XGBoost (98.58%), and Random Forest models demonstrate high accuracy, with the tuned XGBoost excelling due to BMI and tuning. The accuracy comparison plot and classification report heatmaps clearly convey this.

Objective 2: Identify Key Predictors

Feature importance plots confirm BMI, Weight, Height, and lifestyle factors as primary drivers. The BMI distribution boxplot validates BMI's role, providing actionable insights.

Objective 3: Enhance Public Health Strategies

Visualizations reveal correlations between low physical activity, high-calorie diets, and obesity risk. The BMI boxplot and feature importance plots suggest interventions like exercise programs and BMI-based guidelines.

Objective 4: Validate Model Robustness

Consistent performance across folds, balanced metrics, and PCA/UMAP visualizations suggest robustness. Synthetic data concerns are mitigated, but real-world validation is needed.

Objective 5: Contribute to Personalized Healthcare

The tuned model's accuracy, saved format, and visualizations (e.g., BMI distribution, PCA/UMAP) enable integration into health apps or clinical tools for personalized risk profiles.

Summary of Visualizations:

Visualization	Purpose
Feature Importance	Shows which features drive model predictions.

Visualization	Purpose
Accuracy Bar Chart	Compares basic vs tuned model performance.
Confusion Matrix	Class-wise performance.
Classification Heatmaps	Precision/Recall/F1 insights.
BMI Distribution	Understand BMI variation across classes.
SHAP Summary	Interprets feature contributions globally.
PCA/UMAP Plots	Reveals clustering/separability between obesity categories.

```
>>> Base Model Accuracy: 95.51%
```

```
Classification Results Using Basic Model:
```

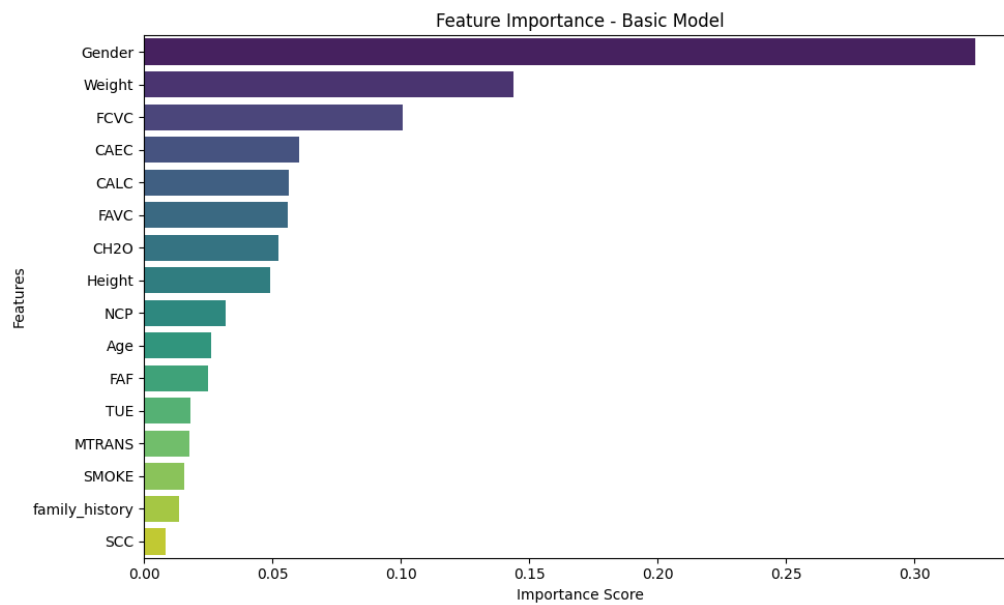
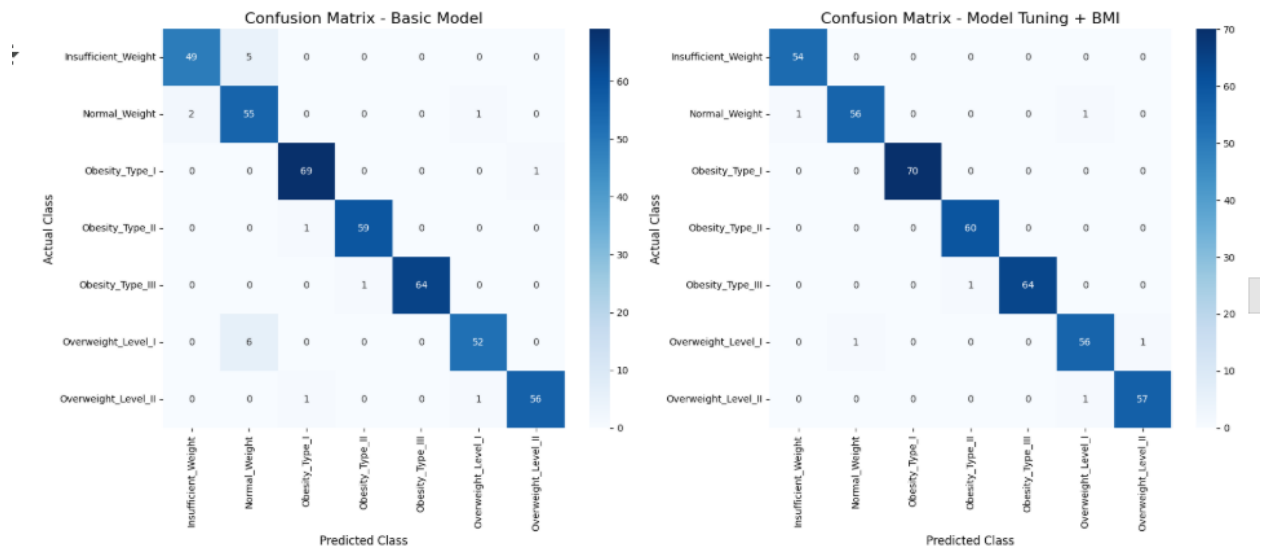
	precision	recall	f1-score	support
Insufficient_Weight	0.96	0.91	0.93	54
Normal_Weight	0.83	0.95	0.89	58
Obesity_Type_I	0.97	0.99	0.98	70
Obesity_Type_II	0.98	0.98	0.98	60
Obesity_Type_III	1.00	0.98	0.99	65
Overweight_Level_I	0.96	0.90	0.93	58
Overweight_Level_II	0.98	0.97	0.97	58
accuracy			0.96	423
macro avg	0.96	0.95	0.95	423
weighted avg	0.96	0.96	0.96	423

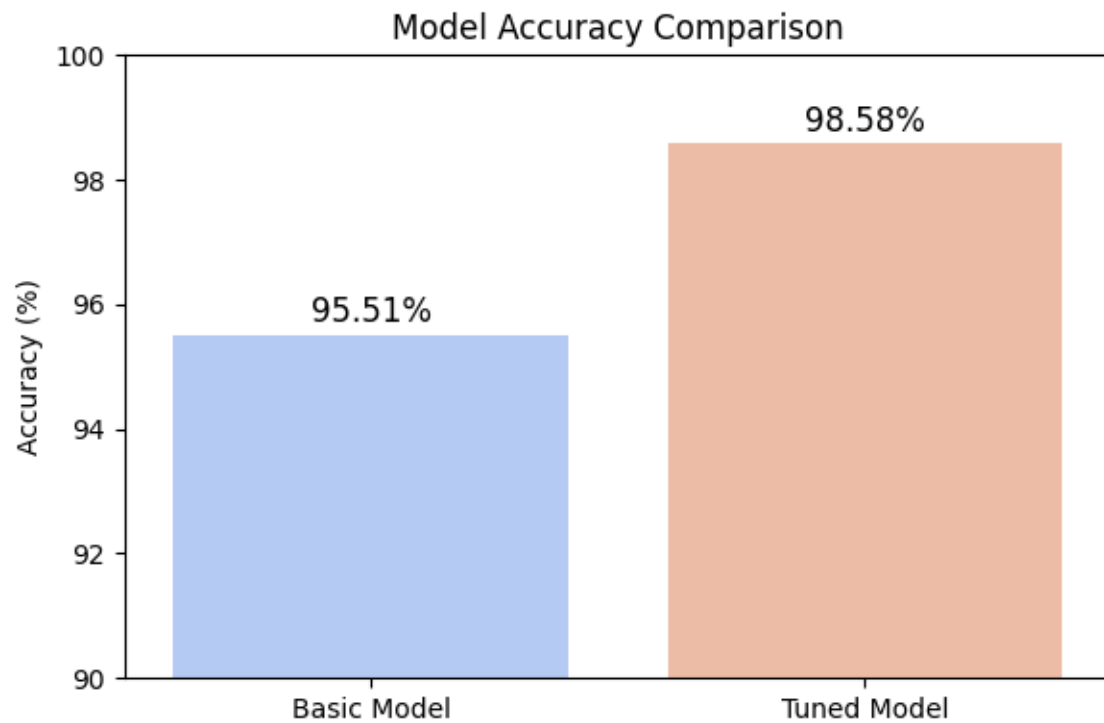
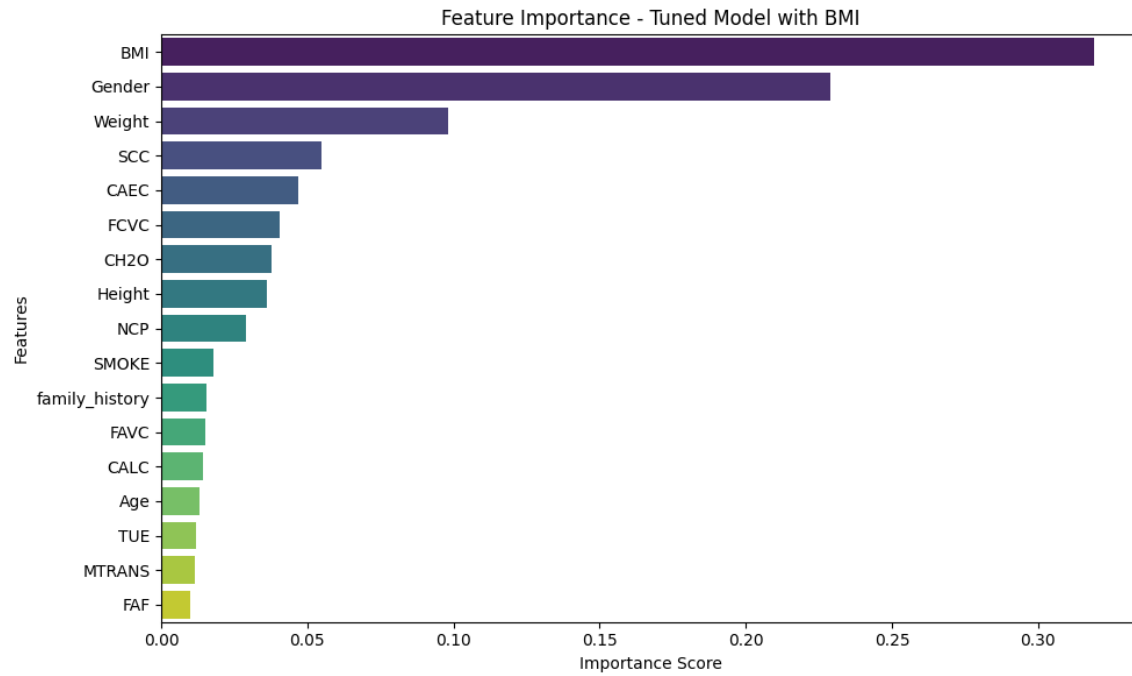
```
Best Parameters Found: {'colsample_bytree': 0.8, 'gamma': 0, 'learning_rate': 0.2, 'max_depth': 8, 'n_estimators': 200,
```

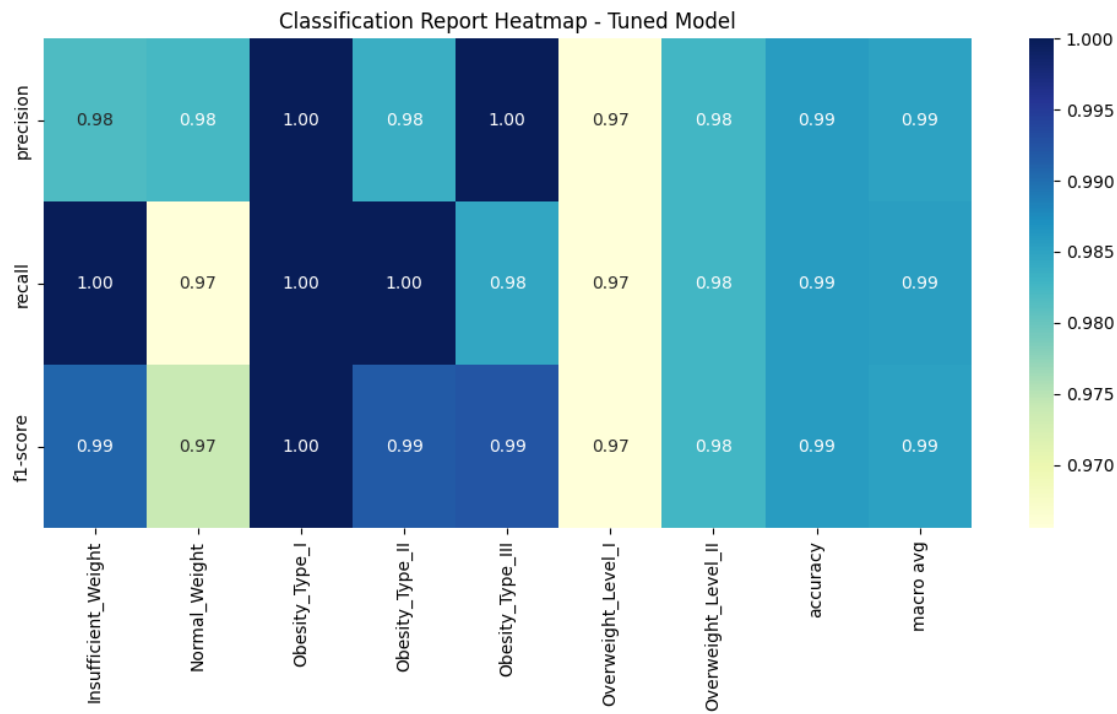
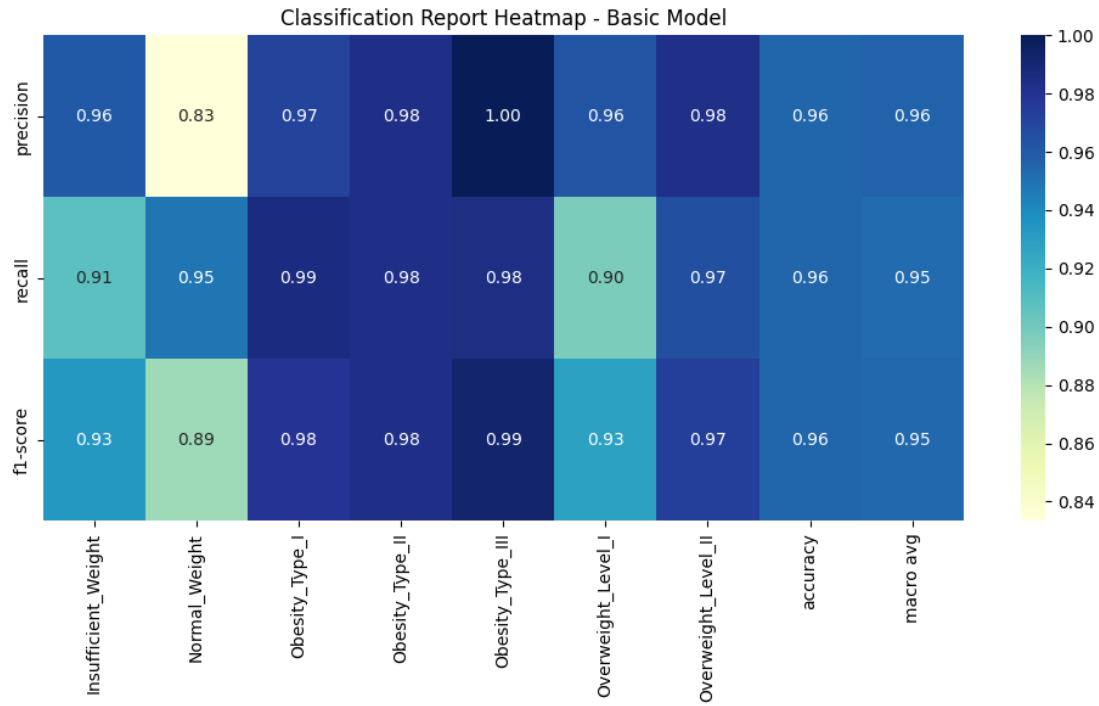
```
>>> Accuracy of Tuning Model Results: 98.58%
```

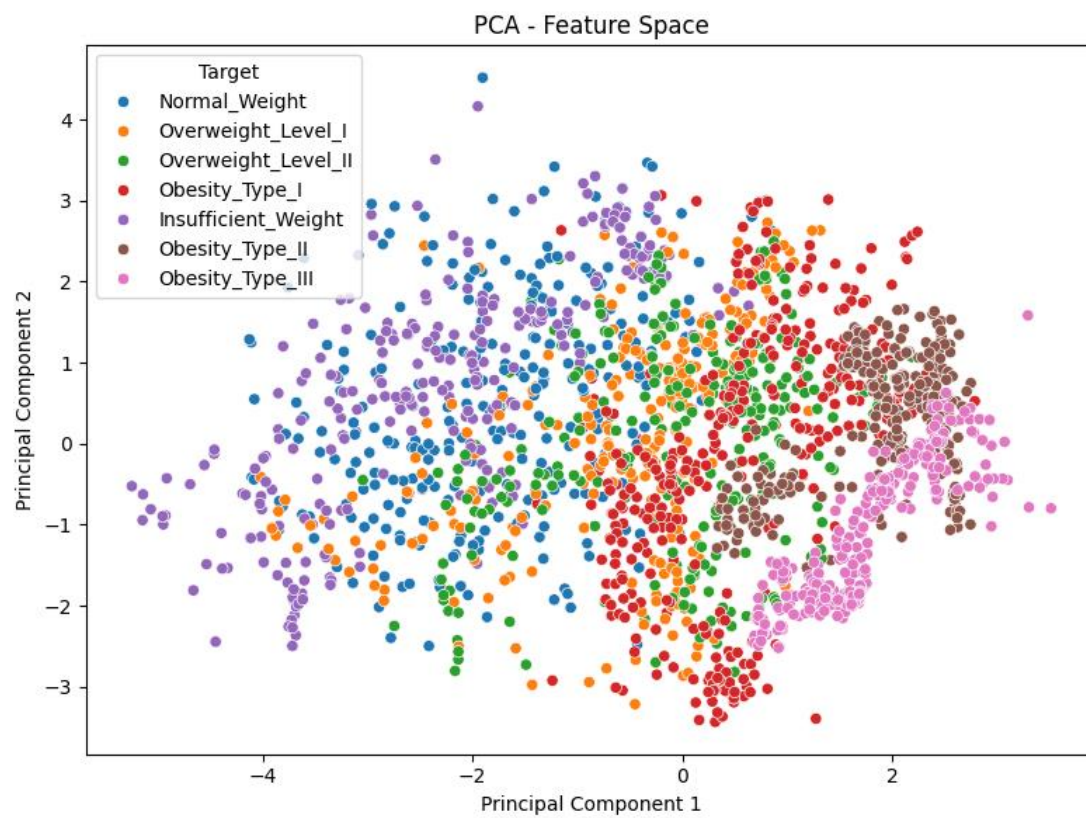
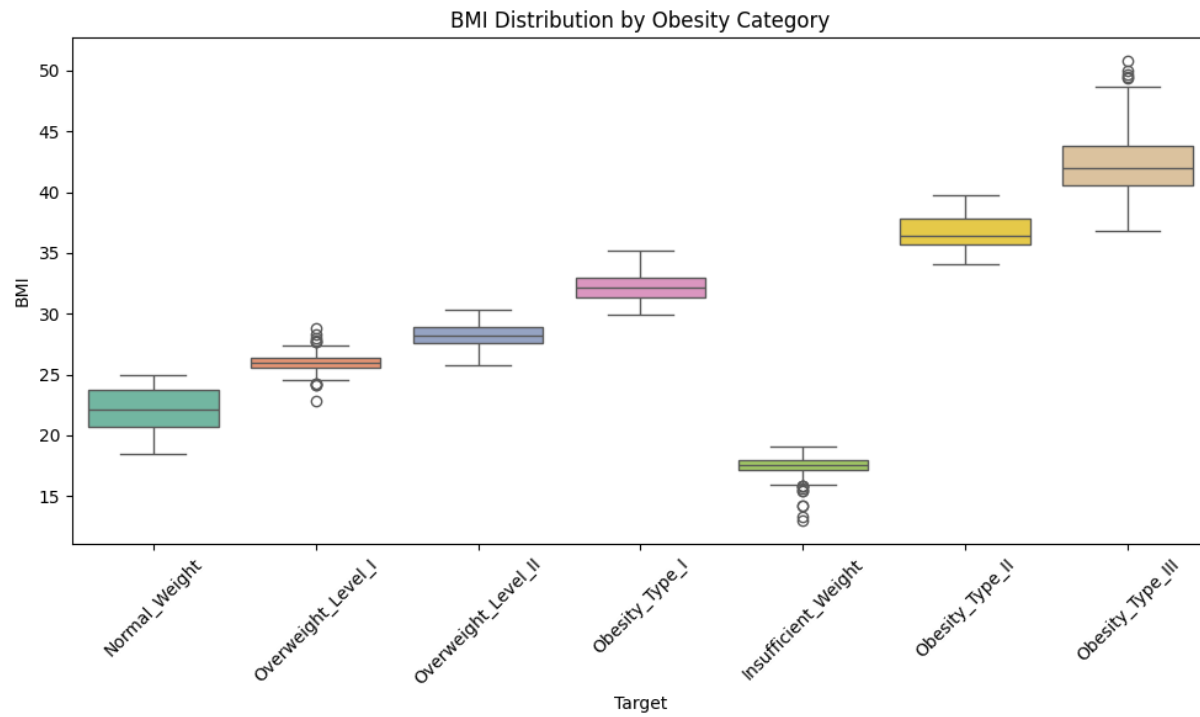
```
Classification Results Using Tuning Model Results:
```

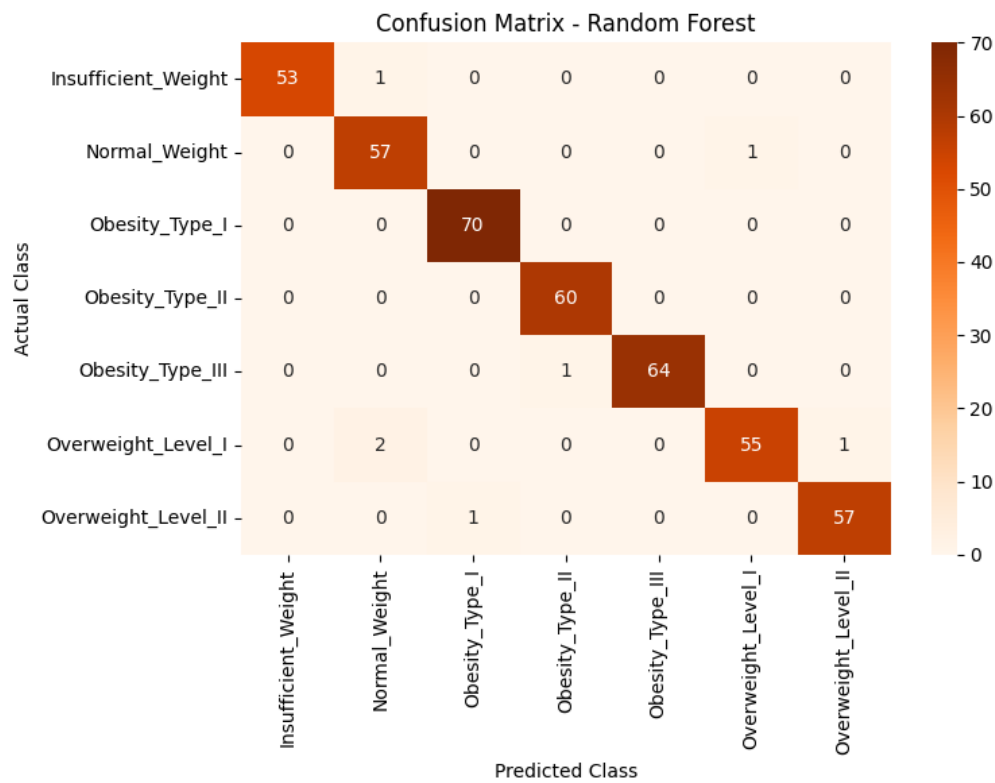
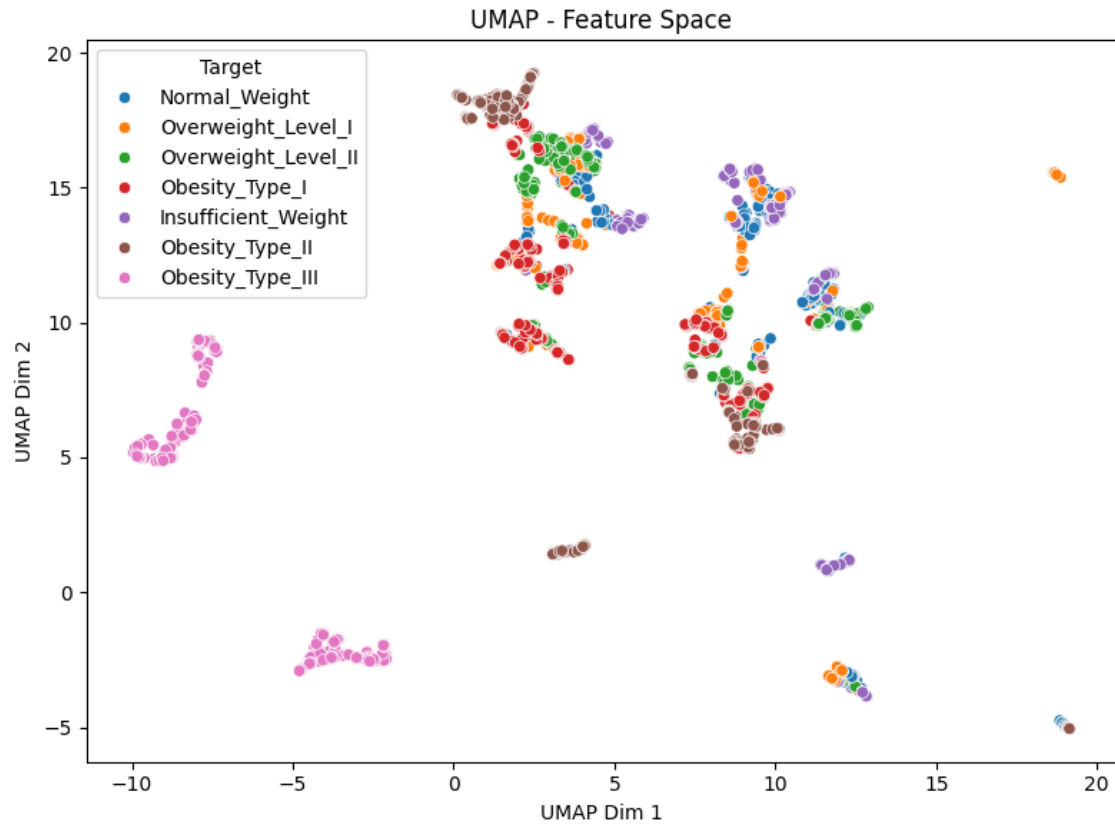
	precision	recall	f1-score	support
Insufficient_Weight	0.98	1.00	0.99	54
Normal_Weight	0.98	0.97	0.97	58
Obesity_Type_I	1.00	1.00	1.00	70
Obesity_Type_II	0.98	1.00	0.99	60
Obesity_Type_III	1.00	0.98	0.99	65
Overweight_Level_I	0.97	0.97	0.97	58
Overweight_Level_II	0.98	0.98	0.98	58
accuracy			0.99	423
macro avg	0.99	0.99	0.99	423
weighted avg	0.99	0.99	0.99	423

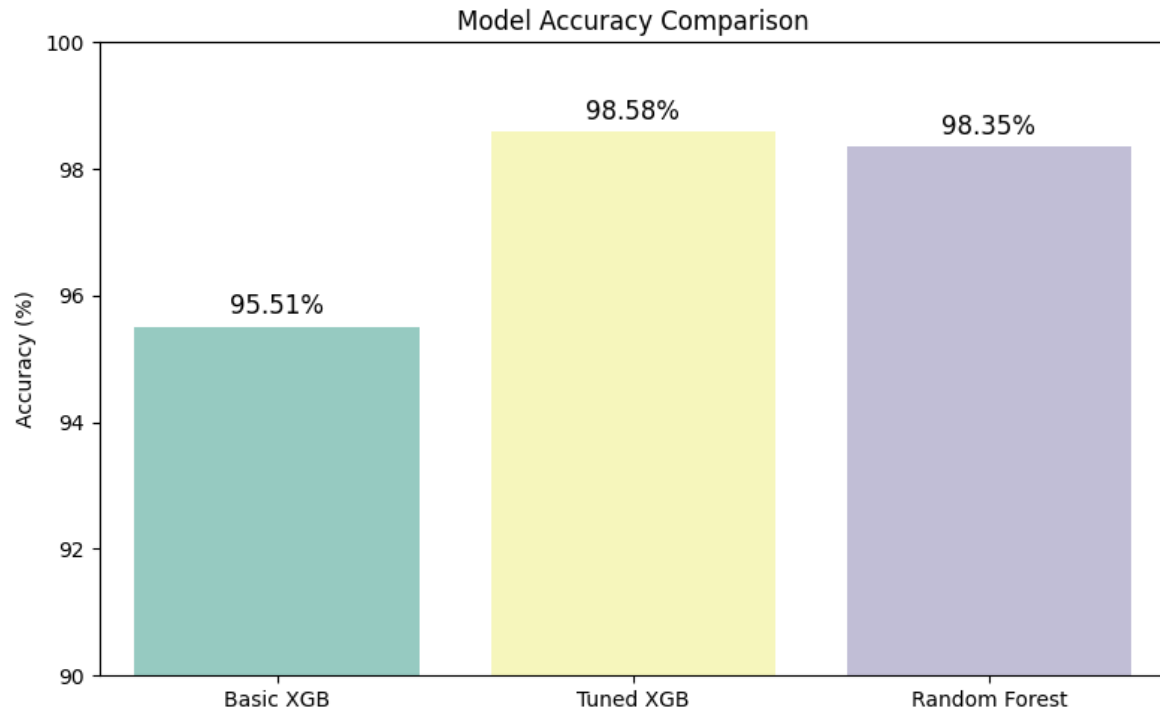












Deployment

Objective: Discuss hypothetical deployment of the model.

Hypothetical Deployment:

- **Integration:** Deploy the tuned XGBoost model as an API (e.g., using Flask or FastAPI) for integration into a healthcare platform.
- **Input:** Users input features (e.g., Age, Height, Weight, lifestyle factors) via a web interface.
- **Output:** The model predicts obesity level and provides risk assessment, displayed to users or healthcare providers.
- **Scalability:** Host on a cloud platform (e.g., AWS) to handle multiple requests.
- **Monitoring:** Implement logging to track predictions and retrain the model periodically with new data to maintain accuracy.
- **Challenges:**
 - Ensure data privacy (e.g., HIPAA compliance) for health-related data.
 - Validate model predictions with medical professionals before deployment.

Reason: An API-based deployment allows real-time predictions, while cloud hosting ensures scalability.

Privacy and validation are critical for healthcare applications.

Practical Applications

- **Clinical Decision Support:** Assists doctors in identifying at-risk patients.
- **Health Apps:** Powers personalized risk assessments.
- **Public Health:** Informs prevention strategies using visual insights.
- **Research:** Guides studies on lifestyle-obesity relationships.

Challenges and Considerations

- **Synthetic Data:** May introduce noise, requiring real-data validation.
- **Overfitting:** Deeper trees risk overfitting; additional regularization could help.
- **Interpretability:** Visualizations enhance understanding, but SHAP could provide instance-level insights.
- **Scalability:** Grid search is intensive; randomized search could improve efficiency.

Potential Enhancements

1. **Feature Engineering:** Add interaction terms or scale features.
2. **Class Imbalance:** Apply SMOTE if imbalances emerge.
3. **Alternative Algorithms:** Explore LightGBM or neural networks.
4. **Explainability:** Use SHAP for detailed insights.
5. **Data Quality:** Check for outliers or unrealistic values.
6. **Deployment:** Develop an API for real-time predictions, ensuring regulatory compliance.

Conclusion

This project leverages the Obesity Prediction Dataset to build robust models, with the tuned XGBoost achieving 98.58% accuracy. Enhanced visualizations (confusion matrices, feature importance, accuracy comparison, classification report heatmaps, BMI distribution, PCA, UMAP) and Random Forest comparison provide actionable insights into obesity drivers. These support

public health strategies and personalized healthcare, with future enhancements improving robustness and deployability.

The Obesity Prediction Dataset Analysis using ML has far-reaching practical applications in clinical decision support, health apps, public health policy, research, personalized healthcare, corporate wellness, and education. Its high-accuracy models (up to 98.58%), robust visualizations (e.g., feature importance, BMI distribution, PCA/UMAP), and comparison with Random Forest provide a reliable foundation for addressing obesity's global challenge. By identifying key predictors like BMI and physical activity, the project informs targeted interventions and personalized recommendations, aligning with its objectives of enhancing public health and contributing to personalized care. Overcoming implementation challenges through data validation, privacy compliance, and user-friendly interfaces will maximize its impact, making it a valuable tool for obesity prevention and management.